KOWSHIN V

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CAREER OBJECTIVE

Aspiring VLSI and Embedded Systems Engineer with hands-on experience in RTL Design, ASIC Design Flow, and Embedded Programming. Seeking opportunities to develop innovative semiconductor and embedded solutions.

EDUCATION QUALIFICATION

BE ECE - Hindusthan College of Engineering and Technology, Coimbatore 2023 - 2027
CGPA: 8.30/10 (Up to VI Semester)

HSE - Saraswathi Matric Hr. Sec. School, Mayiladudhurai 2021 - 2023
Percentage: 81.5%

SSLC - Saraswathi Matric Hr. Sec. School, Mayiladudhurai 2020 - 2021
Percentage: Pass

TECHNICAL SKILLS

VLSI Design

- Verilog HDL (RTL Design)
- Digital Design
- ASIC Design Flow (RTL-to-GDSII)
- RTL Verification (Testbench & Waveform Analysis)
- SpyGlass (Lint, CDC & RDC)
- RTL Synthesis & Physical Design
- Timing Analysis: LEC & GLS

EDA Tools: Synopsys (SpyGlass, VCS, Design Compiler), Cadence (Genus, Innovus, SimVision, Conformal), Xilinx Vivado

Embedded Systems and IoT

- Embedded C Programming
- ESP32 & Arduino Programming
- Sensor Interfacing
- GPIO, UART, I2C & SPI Communication
- IoT Fundamentals

Programming Languages: C Programming, Embedded C, Python (Basic)

INTERNSHIP EXPERIENCE

Angstromers Technologies Bangalore - VLSI Design Engineer Jul 2026 – Present

- Designed and verified an Approximate Tensor Processing Unit (APTPU) using Verilog HDL.
- Performed RTL verification, Lint, CDC & RDC analysis using Cadence SimVision and Synopsys SpyGlass.
- Executed RTL synthesis using Cadence Genus and Physical Design using Cadence Innovus.
- Performed LEC, Gate-Level Simulation (GLS) and timing verification for ASIC signoff.

NLC India Limited - Electronics & Control Department Aug 2025

- Studied industrial instrumentation and control systems used in thermal power plants.
- Observed SCADA and DCS-based automation systems.
- Gained practical exposure to industrial electronic control and monitoring systems.

CERTIFICATIONS (NPTEL ELITE - ONLINE)

Jul 2024 - Oct 2025

- Fundamental of AI: AI basics, search algorithms, reasoning techniques, and introductory machine learning.
- Cloud Computing: Storage, virtualization, and cloud-based IoT application deployment.
- IoT: Sensor integration, device communication, and real-time data monitoring.

PROJECTS

Approximate Tensor Processing Unit (APTPU) – RTL-to-GDSII Tapeout

- Designed and implemented a 12×12 Approximate Processing Element (APE) Systolic Array using Verilog HDL.
- Successfully completed the ASIC RTL-to-GDSII flow, including RTL Verification, SpyGlass (Lint, CDC & RDC), RTL Synthesis, Physical Design, LEC, GLS, and a timing-clean, tapeout-ready implementation.

UART Transmitter and Receiver using Verilog HDL

- Designed and implemented UART Transmitter and Receiver using Verilog HDL with FSM-based RTL architecture.
- Performed functional simulation and testbench verification using Synopsys and Cadence tools.

Smart Geo-Fence Alert for Fisherman

- Developed a GPS-based geo-fencing system to detect maritime boundary crossings and enhance fisherman safety.
- Integrated GSM alerts and buzzer notifications for real-time emergency communication.

ACHIEVEMENTS

Winner of the IVTon – Urbanisation & Smart Cities Track at graVITas 2025, VIT Vellore, for developing an innovative smart city solution. This achievement demonstrates creativity, technical excellence, problem-solving ability, and effective teamwork in addressing real-world urban challenges.

SOFT SKILLS

Problem Solving, Analytical Thinking, Teamwork, Communication Skills, Time Management, Adaptability, Quick Learning

PERSONAL DETAILS

- Date of Birth: 01 October 2005
- Languages Known: Tamil (Native), English (Professional)
- Address: Mayiladuthurai, Tamilnadu